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SUPERSEDING
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22 August 1950

FEDERAL SPECIFICATION

TUBES, STEEL AND OPEN HEARTH IRON, SEAMLESS AND WELDED, BOILER USE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers seamless and electric-resistance-welded boiler tubes, boiler flues, superheater tubes, superheater flues, safe ends and arch tubes.

1.2 Classification.—The boiler tubes covered by this specification shall be furnished in the following compositions:

Composition A.—Steel.
Composition B.—Open hearth iron.

Composition A tubes shall be furnished unless composition B tubes are specified (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

PPP-B-601—Boxes, Wood, Cleated-Plywood.

Military Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-B-121—Barrier-Materials, Greaseproof.
JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier Materials, Waterproof, Flexible.
MIL-C-132—Crates, Open Wood; Maximum Capacity 2,500 Pounds.
MIL-C-16173—Corrosion Preventive, Solvent Cutback, Cold Application.

Federal Standard:

Fed. Std. No. 48—Tolerances for Steel and Iron Wrought Products.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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3. REQUIREMENTS

3.1 Material.—The material for the composition A tubes shall be made by either or both the open-hearth or electric-furnace process. Material for composition B tubes shall be made by the open hearth process.

3.2 Chemical composition.—The chemical composition of the tubes (see 4.3.6) shall conform to table I.

TABLE I.—*Chemical composition*

	Composition A	Composition B
	Percent	*Percent*
Carbon	0.06 to 0.18	Not over 0.03
Manganese	0.27 to 0.63	Not over .03
Phosphorus	Not over 0.048[1]	Not over .020
Sulphur	Not over .058[1]	Not over .045

[1] In the case of electric-resistance-welded tubes, only the following limits for phosphorus and sulphur are permitted:

Phosphorus, maximum percent	0.050
Sulfur, maximum percent	.060

3.3 Fabrication.—Unless otherwise specified (see 6.2), seamless or electric-resistance-welded hot-finished or cold-finished tubes will be acceptable (see 3.6).

3.4 Dimensions.—The dimensions of tubes shall be as shown in table II for the dimensions specified (see 6.1), and shall be within the tolerances specified in 3.5.

3.5 Tolerances.—The dimensions of the boiler tubes offered for delivery shall not vary from the ordered dimensions by amounts greater than those specified in the following referenced paragraphs of Federal Standard No. 48:

Dimension	Paragraph
Outside diameter	16ab1
Wall thickness	16ab2
Length	16ab3
Weight	16ab4
Eccentricity	(See 3.5.1 herein)

3.5.1 *Eccentricity.*—Wall thickness in any one cross section shall not vary more than plus or minus 12½ percent from the actual average wall section. Actual average wall is defined as the average of the thickest and thinnest wall in the section.

TABLE II.—*Standard weights of steel boiler tubes in pounds per foot of length*

Size (outside diameter)	Wall thickness, inch												
	0.083	0.095	0.105	0.109	0.120	0.135	0.150	0.165	0.180	0.200	0.220	0.250	0.280
Inches	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
1¼	.....	1.299	.....	.....	1.608	.....	.....	.....	.....	.....	.....	.....	.....
1½	[1]1.256	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....
2	[1]1.699	1.933	2.125	.....	2.409	2.689	2.964	3.234	3.499	3.845	.....	.....	.....
2¼	.....	2.186	2.405	[2]2.492	2.730	3.049	3.364	3.674	3.979	4.379	4.770	.....	.....
2½	.....	[2]2.440	2.686	[2]2.783	3.050	3.410	3.765	4.115	4.460	4.913	.....	.....	.....
2¾	.....	.....	2.966	.....	3.371	3.770	4.165	4.555	4.941	5.447	.....	.....	.....
3	.....	.....	3.246	[3]3.365	3.691	4.131	4.566	4.996	5.421	5.981	.....	.....	.....
3¼	.....	.....	.....	.....	4.011	4.491	4.966	5.436	5.902	6.515	.....	[1]8.010	.....
3½	.....	.....	.....	.....	4.332	4.852	5.367	5.877	6.382	7.049	.....	.....	.....
3¾	.....	.....	.....	.....	4.652	5.212	5.767	6.317	6.863	7.583	.....	.....	.....
4	.....	.....	.....	.....	.....	5.573	6.168	6.758	7.344	8.117	8.881	.....	.....
4½	.....	.....	.....	.....	.....	6.293	6.969	7.639	8.305	9.185	.....	.....	12.620
5	.....	.....	.....	.....	.....	7.014	7.770	8.520	9.266	10.25	.....	.....	.....
5¼	.....	.....	.....	.....	.....	7.375	8.170	8.961	9.747	10.79	.....	.....	.....
5½	.....	.....	.....	.....	.....	7.735	8.571	9.401	10.23	11.32	.....	.....	.....

[1] Non-standard weight boiler tube.

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3.6 Heat treatment.—Subsequent to the last hot or cold working operation, all electric-resistance-welded tubes shall be normalized by heating to a temperature of at least 1650° Fahrenheit (F.) and cooled in still air without the application of a liquid quench or air blast. When tested as specified in 4.3.7, there shall be no evidence of non-uniformity of structure. Cold-drawn seamless tubes shall be heat-treated as the manufacturer desires as a final process to meet the requirements of this specification except that the minimum heat-treatment temperature shall be 1200°F.

3.7 Hydrostatic pressure.—Each tube shall withstand an internal hydrostatic pressure as specified in 4.3.1 and 4.3.1.1, without showing any leaks, bulges or other defects.

3.8 Soundness.—The tubes shall be of sound structure and capable of passing the nondestructive electrical test specified in 4.3.2.

3.9 Flanging. — Coupons representing the tubes shall be capable of meeting the flange test without cracking or showing flaws as specified in 4.3.4.

3.10 Flattening.—Coupons representing the tubes shall be capable of meeting the flattening test without showing any surface defects as specified in 4.3.5.

3.11 Microscopic examination.—A microscopic examination may be made of electric-resistance-welded tubes to determine that they have been heat-treated properly. (See 4.3.7.)

3.12 Tube marking.—The following markings shall be clearly and legibly painted on each tube 1¼ inches in outside diameter and over provided the length is not under 3 feet as indicated: (a) the name of manufacturer or the trade-mark or brand name by which he can be readily identified; (b) the letters "S" or "ERW" according to process of manufacture; (c) the symbol "WW-T-731", except that seamless tubes taken from mill stock which fully meet the requirements of this specification may bear the symbol "ASTM A 83" or electric-resistance-welded tubes taken from mill stock which fully meet the requirements of this specification may bear the symbol "ASTM A 178" in lieu of "WW-T-731"; (d) suitable marks on each electric-resistance-welded tube tested by the electrical method to show clearly that it was tested by this method (see 4.3.2); and (e) the applicable letter required by (b) shall be followed by "-S" in the case of steel tubes and "-I" for open hearth iron tubes to designate the composition of the tube material. Paint applied to tubes in stencilling shall not be affected by solvents used in subsequent preservation operations.

3.13 Tags.—Tubes less than 1¼ inches in diameter and under 3 feet in length shall have the markings specified in 3.12 clearly shown on a tag securely fastened to each piece or bundle.

3.14 Workmanship. — Finished tubes shall be reasonably straight, true to form, free from injurious defects (laminations, cracks, seams, tears), shall have smooth ends free from burrs, and shall have a workmanlike finish.

3.14.1 The outside of electric-resistance welded tubes shall be free from protruding flash and the flash on the inside shall not protrude more than 0.010 inch at any point in the tube. Welded tubes shall have a clean smooth weld and the increase in wall thickness at the weld shall not be more than 20 percent of the ordered thickness.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—For inspection purposes a lot shall consist of 250 tubes or less, or 2,000 feet or less, of safe end material of the same type, composition, and size.

4.2 Visual surface and dimensional inspection.—All tubes shall be visual surface inspected for defects by rolling on a set of "horses" or other suitable supports, so that the entire surface of each tube may be properly inspected. The diameter, wall thickness and length shall be measured. The thickness of walls of tubes shall

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be determined by a suitable micrometer or gage at the option of the procuring activity. Any tube failing to pass the surface inspection and dimensional tests shall be rejected. In order to determine compliance with weight requirements the tubes shall be weighed in lots under the supervision of the Government inspector. However, the Government inspector may weigh individual tubes when the dimensions indicate that the permissible weight has been exceeded.

4.3 Tests.

4.3.1 *Hydrostatic test.*—Each tube shall be tested at an internal hydrostatic pressure of 1,000 pounds per square inch.

4.3.1.1 Each welded tube shall withstand being struck near each end while under test pressure with a 2-pound hammer or the equivalent. Individual tubes which show any defect under this test shall be rejected.

4.3.2 *Alternate electrical test for electric-resistance-welded tubes (at the place of manufacture).*—When specified (see 6.2), each electric-resistance-welded tube shall be subjected to a non-destructive electrical test in lieu of the hydrostatic test (see 4.3.1 and 4.3.1.1). The test shall be capable of detecting all injurious defects or imperfections, interior or exterior, having a length of 1/16 inch or greater and a radial depth equal to or greater than one-half of the wall thickness, and any defects of any length completely penetrating the wall which are in the weld seam or the adjacent metal affected by the weld. Individual tubes which show any defects under this test shall be rejected.

4.3.3 *Samples for flange and flattening test.*

4.3.3.1 *Flange test.*—For seamless boiler tubes having wall thicknesses (actual mean wall) less than 10 percent of their outside diameter (and providing the thickness does not exceed 0.200 inch) and for all sizes of electric-resistance-welded boiler tubes, two tubes shall be selected by the Government inspector from each lot of material for the test specified in 4.3.4.

4.3.3.2 *Flattening test.*—Two tubes shall be selected by the Government inspector from each lot of material for the test specified in 4.3.5.

4.3.3.3 *Test specimens.*—Test specimens required for the flange and flattening tests shall be taken from each end of finished tubes and, when practicable, before being cut to the required length. Specimens shall be smooth on the ends and free from burrs.

4.3.4 *Flange test.*—A test specimen not less than 4 inches in length from each tube taken for test shall have a flange turned over at right angles to the body of the tube without cracking or showing any flaws. This flange, as measured from the outside of the tube, shall in no case be less than 1/8 inch nor more than 1/2 inch. Within these limits the width of the flange shall be as shown in table III. The flange test shall be conducted at room temperature. If both tubes (see 4.3.3) fail this test, the lot shall be rejected. If one tube fails, retests shall be made as specified in 4.3.4.2.

TABLE III.—*Width of flange*

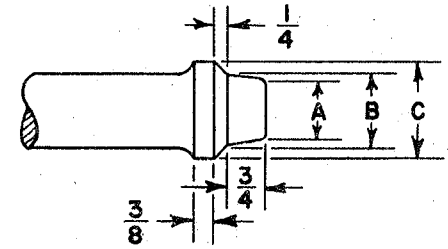
Outside diameter of tube, inches	Width of flange
	Composition A and composition B
2½ and under	15 percent of outside diameter
Over 2½ to 3¾, incl.	⅜ inch[1]
Over 3¾	10 percent of outside diameter

[1] 12.5 percent of outside diameter for electric resistance welded tubes.

4.3.4.1 *Flange tool.*—The tool and die block shown on figure 1 may be used in making the flange test.

4.3.4.2 *Retests.*—If one of the two tubes tested does not conform to 4.3.4, a retest of two additional tubes from the same lot may be made. Each of these tubes shall conform to the requirements specified herein. If either retest tube does not meet the requirements, the lot shall be rejected.

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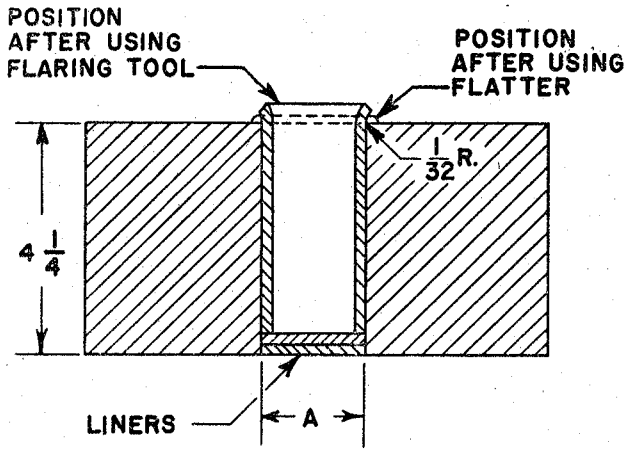


A= O.D. OF TUBE LESS 5/8

B=O.D. OF TUBE LESS 3/8

C=O.D. OF TUBE PLUS 3/16

FLARING TOOL



A = O.D. OF TUBE PLUS 1/32

DIE BLOCK

FIGURE 1.—Flaring tool and die block for flange test.

4.3.5 *Flattening test.*—A test specimen, not less than 2½ inches in length from each sample tube taken for test shall withstand flattening cold (room temperature) between parallel plates as specified in 4.3.5.1. If both tube specimens fail the flattening test, the lot shall be rejected. If one specimen fails, a retest may be made (see 4.3.5.2).

4.3.5.1 There shall be no evidence of lamination, or burnt material, or incomplete penetration of the weld (in the case of welded tubes) until the distance between the plates is as follows:

(a) Three times the wall thickness for seamless tubes.

(b) Three times the wall thickness for electric-resistance-welded tubes with the weld located 90 degrees from the line of direction of the applied force.

No cracks or breaks in the metal shall occur in flattening to the distance shown in table IV. Superficial ruptures as a result of minor surface imperfections shall not be cause for rejection.

4.3.5.2 *Retests.*—If one of the two tubes tested does not conform to 4.3.5, retests of two additional tubes from the same lot shall be made. Each of these tubes shall conform to the requirements specified herein. If either retest tube fails, the lot shall be rejected.

4.3.6 *Chemical analysis.*—Chemical analysis may be made by the Government inspector or through him by any Government laboratory or other designated representatives without cost to the contractor.

4.3.6.1 *Sample.*—Drillings, millings, or clippings for chemical analysis shall be taken from

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TABLE IV.—*Distance between plates*[1]

Outside diameter	Distance between plates												
	0.083 inch wall	0.095 inch wall	0.105 inch wall	0.109 inch wall	0.120 inch wall	0.135 inch wall	0.150 inch wall	0.165 inch wall	0.180 inch wall	0.200 inch wall	0.220 inch wall	0.250 inch wall	0.280 inch wall
Inches	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
1¼	....	0.67	....	....	0.75	....	....	....	....	....	....	....	....
1½	0.68	....	....	....	....	....	....	....	....	....	....	....	....
2	.75	.75	0.80	....	.87	0.94	0.99	1.04	1.09	1.14	....	....	....
2¼	....	.78	.84	0.93	.91	.98	1.04	1.10	1.15	1.21	1.28	....	....
2½	....	.89	.87	.97	.95	1.02	1.09	1.15	1.21	1.28	....	....	....
2¾	....	....	.89	....	.98	1.06	1.13	1.20	1.26	1.34	....	....	....
3	....	....	.91	1.03	1.01	1.09	1.17	1.24	1.31	1.39	....	....	....
3¼	....	....	....	....	1.03	1.12	1.20	1.28	1.35	1.44	....	1.75	....
3½	....	....	....	....	1.05	1.15	1.23	1.31	1.39	1.48	....	....	....
3¾	....	....	....	....	1.07	1.17	1.26	1.34	1.42	1.52	....	....	....
4	....	....	....	....	....	1.19	1.28	1.37	1.45	1.55	1.80	....	....
4½	....	....	....	....	....	1.23	1.32	1.42	1.51	1.62	....	....	2.16
5	....	....	....	....	....	1.26	1.36	1.46	1.56	1.68	....	....	....
5¼	....	....	....	....	....	1.27	1.38	1.48	1.58	1.70	....	....	....
5½	....	....	....	....	....	1.28	1.40	1.50	1.60	1.72	....	....	....

[1]The distance between plates shall be determined by the following formula:

$$H = \frac{(1 + e)\ t}{e + t\text{-}D}$$

Where:

H = distance between flattening plates
e = 0.09 (deformation constant)
t = nominal wall thickness; the minimum wall thickness as ordered divided by 0.875
D = outside diameter

several points around each tube selected by the Government inspector for physical tests and the chips from each tube combined and thoroughly mixed. The resultant sample shall be free from oil, dirt, grit, and other foreign matter and shall consist of at least 1 ounce. If the analysis is not within the limits specified in 3.2, the lot shall be rejected.

4.3.7 *Microscopic examination.*—Microscopic examination of electric-resistance-welded tubes may be made by the Government inspector or through him by any Government laboratory or other designated representative without cost to the contractor.

4.3.7.1 *Rejection.*—The Government inspector may make a microscopic examination of any tube in a lot to determine whether the electric-resistance-welded tubes have been properly heat treated (normalized). Any evidence of non-uniformity of structure shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary requirements.

5.1.1 *Coating.*—After inspection and prior to shipment (see 6.2), all tubes shall be coated inside and outside with a rust-preventative compound in accordance with commercial practice.

5.1.2 *Packing.*—The tubes shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.1.3 *Marking.*—Shipping containers shall be marked with the name of the commodity, the composition, quantity contained therein, name of the manufacturer, and the contract or order number under which delivery is made.

5.2 Military requirements.

5.2.1 *Preservation.* — The tubes shall be coated inside and out with preservative conforming to grade 2 of Military Specification MIL–C–16173.

5.2.2 *Packaging and packing.*

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5.2.2.1 *For domestic shipment, immediate use.*—The tubes shall be prepared for shipment in a manner to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Loading shall be accomplished in a manner to permit maximum utility of mechanical handling equipment in unloading.

5.2.2.2 *For domestic shipment and storage.*—Tubes which are segregated by composition and by outside diameter and length in quantities shown in table V, preserved as specified in 5.2.1 shall be packed in a cleated bundle as shown on figure 2. The tube bundles shall have tension tied steel strapping over nominal 2 by 6 inch wood cleats which are positioned approximately 2 feet from each end of unit load. Unit loads exceeding 10 feet in length shall be provided with an additional cleat band located equidistant from the end cleat bands. Portions of the tube bundle coming in contact with the cleats shall be wrapped with a double thickness of barrier material conforming to type I of Military Specification JAN–B–121. Two parallel girthwise steel straps of coated or galvanized flat steel 1¼-inch by 0.035 inch or equivalent round steel wire shall be tension tied over each cleat band. Galvanized corner protectors shall be required over end grain when round wire strapping is used. Cleats may be staggered as required to permit stacking. The ends of the unit load shall be completely covered with barrier material conforming to type I of Military Specification JAN–B–121 and overwrapped with burlap. The protective end cover shall be secured in place with steel strapping as shown on figure 2.

TABLE V.—*Quantities of tubes in a cleated bundle*

Outside diameter	Length	Rows high	Number of tubes
Inches	*Feet*		
1¼	Up to 20	5	78
1½	Up to 20	5	68
2 and 2¼	Up to 20	4	50
2½ and 2¾ ...	Up to 20	4	30
3 and over	All lengths	3	20
1¼, 1½ and 2..	Over 20	5	50 to 75 approx.

5.2.2.2.1 When the quantity of tubes of the same composition and outside diameter and length does not permit the fabrication of a cleated bundle containing the quantities shown in table V, the number of tubes contained in the bundle may be lowered by decreasing the width requirement with or without an increase in the height of the bundle.

5.2.2.2.2 Small quantities of tubes which do not permit packing in cleated bundles shall be hand bundled not to exceed approximately 200 pounds gross weight per bundle. The bundle shall be strapped with coated or galvanized flat steel or round wire strapping. End protection shall be as specified in 5.2.2.2.

5.2.2.3 *For overseas shipment.*—Tubes in quantities shown in table V shall be packed in cleated bundles as specified in 5.2.2.2. Prior to fabricating the cleated bundle, the quantity of tubes shall be completely wrapped, including the ends of the load, with grade A greaseproof barrier material conforming to Military Specification JAN–B–121 (not required over ends) and waterproof barrier material conforming to type E2C or E3C of Military Specification JAN–P–125. As alternates to the waterproof barrier material required, a combination of barrier materials conforming to Military Specification JAN–P–125 of two thicknesses of one type may be used providing the total tensile strength of the material is equal to the tensile strength of the specified material (65 pounds per square inch of width) and the outer wrap is a reinforced material (types E1, E2, E3 and L4 of Military Specification JAN–P–125). Additional strapping, as required, shall be used to secure the waterproof wrap. The requirement for end protection specified in 5.2.2.2 shall be required. When the quantity of tubes of the same composition and size does not permit the fabrication of a cleated bundle containing the quantities shown in table V, the tubes shall be packed as specified in 5.2.2.2.1 subject to the additional requirement of this paragraph.

5.2.2.3.1 As an alternate, tubes may be packed in cleated plywood or nailed wood boxes or unsheathed crates conforming to overseas type

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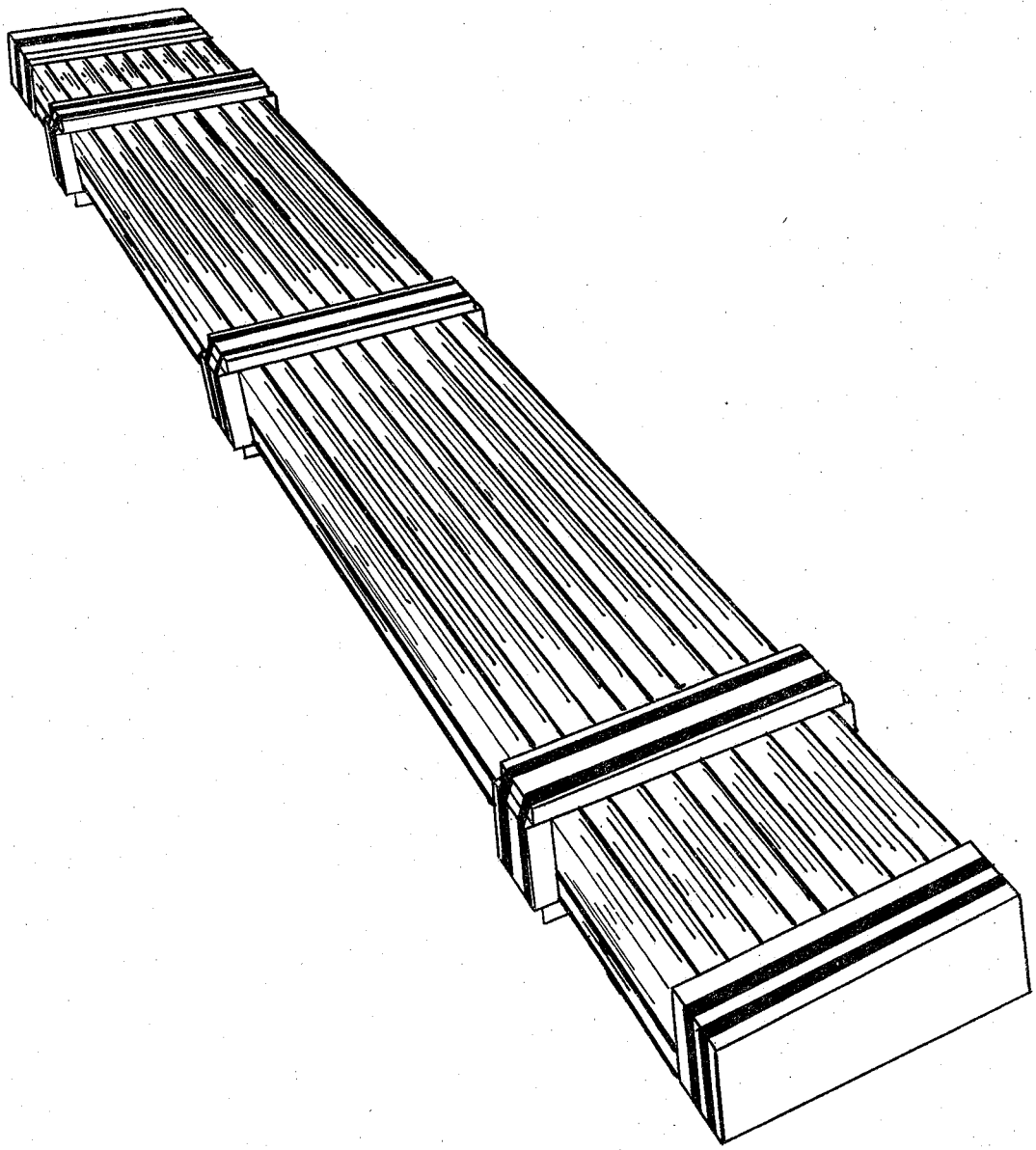


FIGURE 2.—Typical cleated bundle.

of Federal Specification PPP–B–601, Military Specifications JAN–P–106, or style C of MIL–C–132. Prior to packing, the quantity of tubes shall be completely wrapped with grade A barrier material of Military Specification JAN–B–121 and overwrapped with waterproof barrier material conforming to Military Specification JAN–P–125. The maximum length requirement of the crates shall be waived. Boxes shall be modified by the addition of 2 by 4 inch skids positioned perpendicular to the longest dimension. The minimum number of skids required shall be two. The gross weight of the boxes and crates shall not exceed approximately 500 pounds.

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5.2.2.4 *Marking.* — In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. — Tubes specified herein are for use in stationary and locomotive boilers. The tubes are also for use in boilers of auxiliary ships at pressures of 500 pounds per square inch and under where commercial tubes in accordance with designation ASTM A–83 or ASTM A–178 are acceptable.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) If composition B is required (see 1.2).

(c) Type of tubes acceptable (see 3.3).

(d) Dimensions required (see 3.4).

(e) Outside diameter when ring gage is used (see 3.5.1).

(f) When a non-destructive electrical test may be used (see 4.3.2).

(g) Coating required (see 5.1.1 and 5.2.1).

(h) Whether preparation for domestic shipment, immediate use; domestic shipment and storage or overseas shipment is required (see 5.2).

6.3 Major details. — This specification conforms in all major details to that of the 1952 A. S. M. E. Boiler Construction Code, and to A. S. T. M. Specifications A 83–54T and A 178–54T for the sizes and compositions covered.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
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Navy—Sh Y
Air Force.

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